

Designed by Remus Teodorescu 2026

LECTURE 12.2

Power Grid Stability Prediction

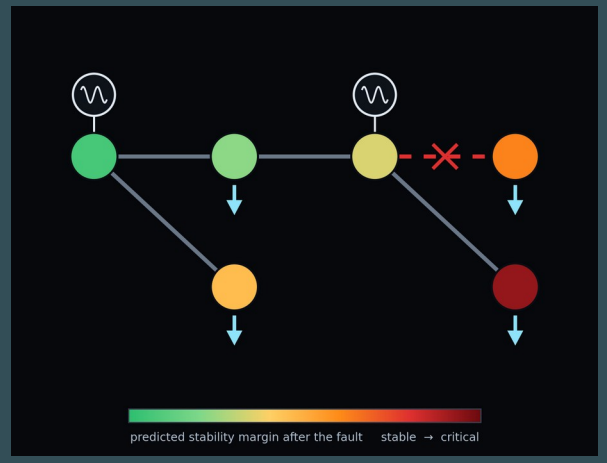
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Deep Learning for Engineering · 2026

FAULT · MARGIN · SCREEN



A line faults and the protection needs 140 ms to clear it: does the plant stay in sync? The answer is the critical clearing time, one simulation per case, and the cases outgrow the time available. The last lecture screens them with a graph network and tests it on a topology it has never seen.

Learning Objectives

SECTIONS, IN ORDER

1. **Transient stability** — the clearing time, the scale, the margin
2. **The graph predictor** — an edge removed, and the honest test
3. **The screen in service** — microseconds, and what stays a simulation

UDL carries the graph network. The example is Ex_12.2's screening of the six-bus grid of the last lecture; the labels are the classical machine model's - no exciter, no governor, first swing only - and results inherit that.

READ, AND WHAT EACH ANSWERS

SciPy	how is a swing integrated?
UDL ch. 13	what does a graph layer share?
PyTorch	how is a model shipped?

The Question a Relay Asks

Critical clearing time — the longest a fault may last before the machines fall out of sync, which happens when the angle between them passes 180 degrees. It belongs to the case: the operating point, the line that is out and where the fault is. A case is insecure when its critical time is 140 ms or less, the time the protection needs to clear a fault.

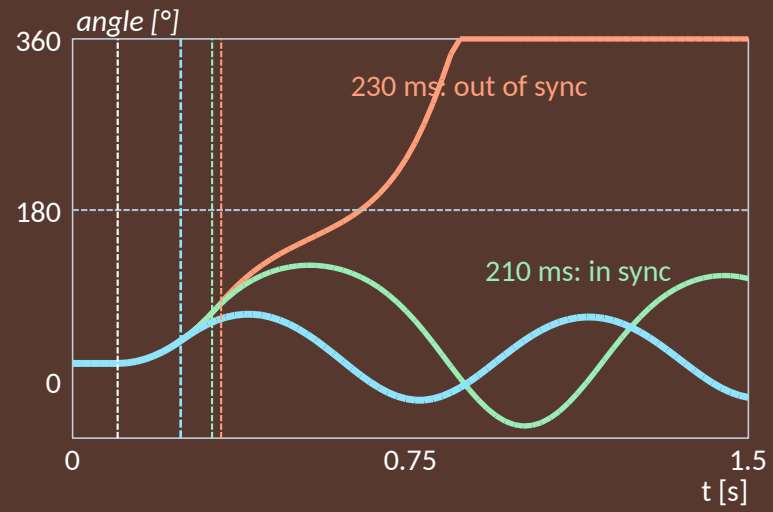
THE LABEL OF EVERY CASE

$$\mathrm{CCT} = \max\left\{ t_c : \max_t |\delta_1(t) - \delta_0(t)| < 180^\circ \right\}, \qquad \text{insecure} \Leftrightarrow \mathrm{CCT} \le 140 \text{ ms}$$

t_c the clearing time, counted from the start of the fault, $\delta_1 - \delta_0$ the angle between the machines
140 ms the time the protection needs to detect the fault and open the breakers

Why a short critical time is insecure: the protection cannot clear a fault in less than 140 ms (30 ms for the relay to decide, 50 ms for the breaker to open, 60 ms of margin). If a case falls out of sync after 120 ms, the breakers open 20 ms too late and the plant must be disconnected. The question is therefore whether the critical time is longer than 140 ms. Of the three stability questions, angle, frequency and voltage, this lecture asks the first.

ONE CASE, ACTUAL AND WHAT-IF



The fault starts at the white line. The protection clears it 140 ms later (cyan): the angle peaks at 70° and returns. Green and orange are what-ifs, cleared at 210 and 230 ms; the critical time, 220 ms, lies between. 220 > 140, so the case is secure.

Too Many Cases to Simulate

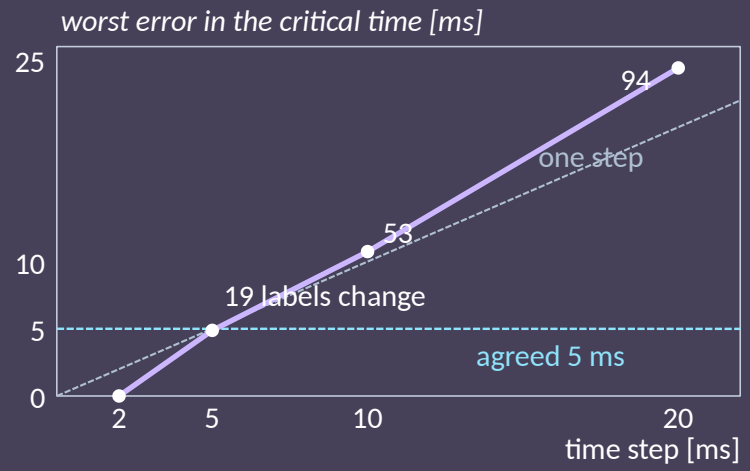
Contingency screening — the operator's standing check of N-1 security: for the present operating point and every contingency - a fault, then the loss of any one line - does the grid stay in sync? Every operating point combined with every contingency is a case, and each case needs about ten simulations in time.

THE COUNT, ON SIX BUSES ALREADY

operating points	180, drawn around the base case
contingencies	6: a fault at the plant, then no line or one line lost
cases	1080, and 16 % of them insecure

How one case is labelled: the swing equations are integrated in time steps of 2 ms, with the fault cleared after a trial time. The trial time is then moved halfway up or down - longer if the machines stayed in sync, shorter if not - until the critical clearing time is known within 2 ms. A real grid has thousands of buses and a state that changes every few minutes, so a cheaper screen is needed, with the simulation kept for the doubtful cases.

THE TIME STEP AND THE LABEL'S ERROR



The time step is how far the simulation advances at a time, not the sampling time of a meter. The fault can only be cleared at the end of a step, so the critical time is known to about one step. 5 ms is the longest step within the agreed 5 ms, three times cheaper than 2 ms, yet 19 of 1080 labels change: many cases lie near 140 ms.

The Equal Area Criterion

Equal area criterion — stability read as energy on the power-angle curve: the machine stays in sync if the energy the grid can take back after clearing is at least the energy the fault gave the rotor. Their difference is a margin: how far a case is from losing sync, not only whether it does.

THE TWO AREAS AS ENERGY

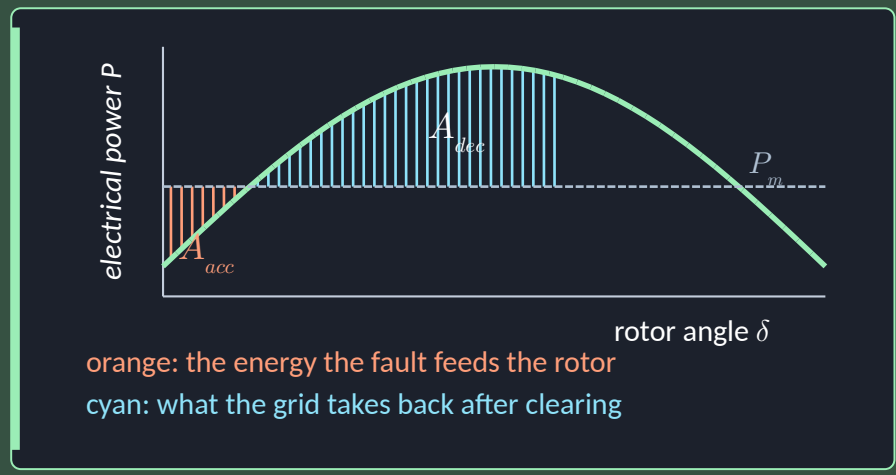
$$A_{acc} = \int_{\delta_0}^{\delta_c} (P_m - P_e^F(\delta))\,d\delta \le A_{dec} = \int_{\delta_c}^{\delta_{max}} (P_e(\delta) - P_m)\,d\delta$$

P_m the mechanical power, P_e^F the electrical power during the fault, P_e after clearing, δ the rotor angle δ_0 before the fault, δ_c at clearing, δ_{max} where P_e falls below P_m again; margin = $A_{dec} - A_{acc}$

In words: during the fault P_e falls below P_m and the rotor speeds up, gaining the energy A_{acc}. After clearing P_e exceeds P_m and the rotor slows, but the grid can take back at most A_{dec} before δ_{max}. Clearing later makes A_{acc} larger and A_{dec} smaller; the critical clearing time is where they are equal.

TALK · 2 MIN How would you explain, in your own words, why an operator wants a margin rather than a yes or no? Then, in the drawing on the right, what happens to the two areas when the fault is cleared later?

THE TWO AREAS, ON THE CURVE



orange: the energy the fault feeds the rotor

cyan: what the grid takes back after clearing

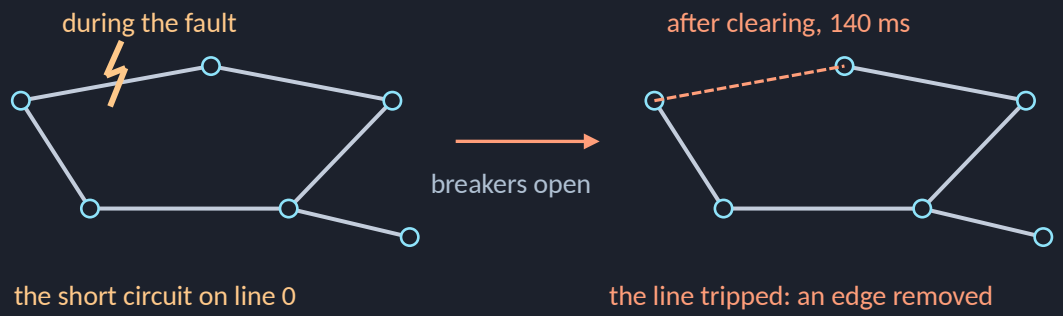
by hand	one machine judged without simulating in time
in Ex_12.2	the margin in time: CCT minus 140 ms
why a margin	the screen can rank cases, nearest to losing sync first

An Edge Removed

Contingency as a graph change — a short circuit on a line - a lightning strike, a falling branch - is cleared by opening the breakers at the line's two ends, so the lasting change is the line's removal: an edge gone from the graph, not a new value in a fixed list of inputs.

THE SAME BUSES, A DIFFERENT GRAPH

LINE 0 OUT, AS THE NETWORK SEES IT



In the language of L5.1: the tripped line is removed from the adjacency matrix A, where A_{ij} = 1 if a line joins buses i and j. Its two buses stop exchanging numbers, and the same weights run on the new graph. Any line can be the one lost.

WHAT GOES IN, WHAT COMES OUT

at each bus	6 numbers: P, Q, $\lvert V\rvert$, θ before the fault, and two flags: machine, fault
the grid	which buses are joined once the line is opened: the adjacency matrix
out	one number, the critical clearing time of the case
weights	6 721, the same at every bus and for every contingency

Which line is out is not one of the numbers at the buses: the network learns it only from the grid it is given.

Message Passing

Message passing — the graph network's one move: every bus combines its own numbers with those of the buses it is joined to, through two weight matrices that are the same at every bus. One layer reaches one line further, so three layers span this grid.

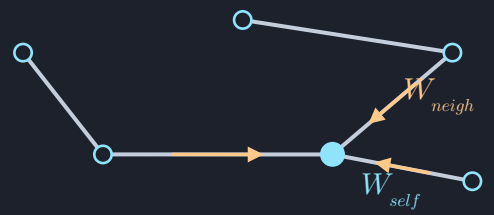
ONE LAYER, AND THE OUTPUT

$$H^{(l+1)} = \tanh\left(H^{(l)} W_{self} + b + \hat{A} H^{(l)} W_{neigh}\right), \qquad \hat{A} = \tilde{D}^{-1/2}(A + I)\tilde{D}^{-1/2}$$

$$\mathrm{CCT}_{pred} = \mathrm{MLP}\left(\frac{1}{B}\sum_{i=1}^{B} h_i^{(3)}\right)$$

$H^{(l)}$ the numbers at the B buses after layer l, one row per bus; W_{self}, W_{neigh}, b weights shared by all buses
A the adjacency after the line is opened, I each bus linked to itself, $\tilde{D}$ its number of links; MLP a small dense network

ONE LAYER, ON THE EXERCISE'S GRID



bus 3 adds its neighbours' numbers to its own

one layer: one line further

the same W_{self}, W_{neigh} at every bus

so any graph fits: the outage included

MEASURED ON 216 TEST CASES

	simulation, 5 ms	network
mean error in the CCT	1.66 ms	0.57 ms
insecure cases missed	0 of 26	0 of 26
secure cases flagged	5	0
time for one case	0.6 ms	3 µs
training, once	—	11 s

The test cases are new operating points with contingencies seen in training, the easier test; a contingency never seen comes two slides on. No dense network is trained, so the notebook makes no claim against one.

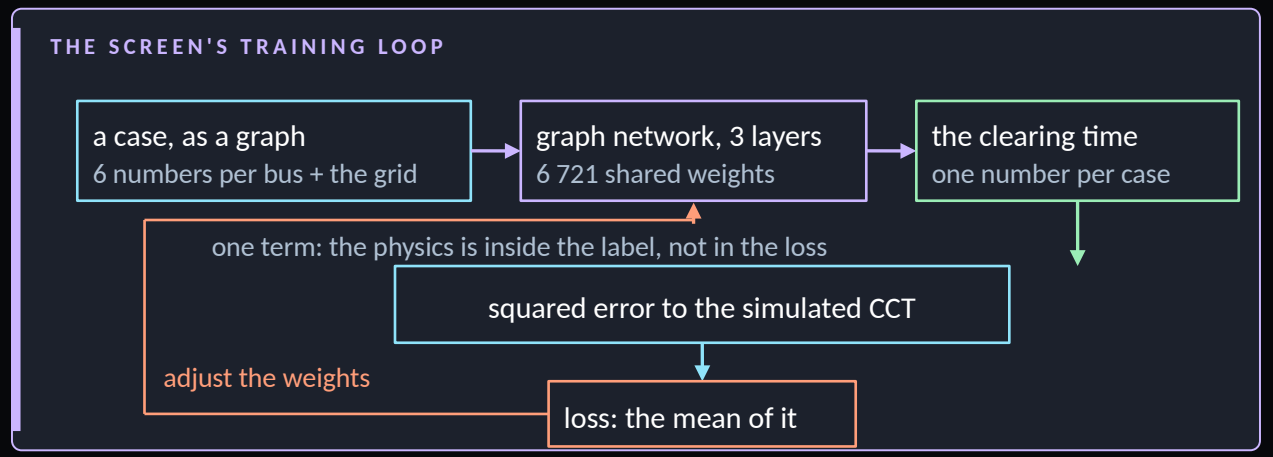
Training the Graph Screen

Supervised training on simulated labels — the screen is trained like L4's regressions, not like L12.1's estimator: the simulator makes the labels - 864 cases, each with its critical clearing time - and the loss is the squared error between the network's time and the simulator's.

THE LOSS AND THE TRAINING LOOP

$$\mathcal{L} = \frac{1}{N}\sum_{n=1}^{N} (\mathrm{CCT}_{pred,n} - \mathrm{CCT}_n)^2, \qquad N = 864$$

$\mathrm{CCT}_{pred,n}$ the network's critical clearing time for case n, CCT_n the simulated one; no swing equation in the loss



THE TRAINING, IN NUMBERS

labels	critical clearing times from the 2 ms simulation
training cases	864: 144 operating points × 6 contingencies
optimiser	Adam for 1 500 steps, then L-BFGS for 40; 11 s
the physics	in the labels and the graph, not in the loss
retrain	when the grid itself changes: a new line or machine

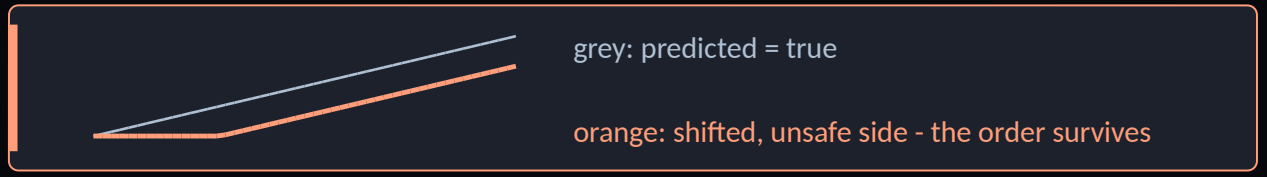
Unlike L12.1, the swing equation is not in the loss: the critical clearing time is the result of about ten simulations, not a quantity the equation fixes at each instant.

Unseen Topologies

The honest test — new operating points with contingencies seen in training test only interpolation. The fair test is a contingency never seen: the network is trained without line 0's outage, then asked about it. The exercise does this and reports what happens.

WHAT THE EXERCISE MEASURES

line 0 out, seen	mean error 0.69 ms, no insecure case missed
line 0 out, unseen	mean error 57 ms, every CCT predicted too long
insecure missed	all 11 of the 11 insecure cases
threshold 185 ms	instead of 140: none missed



grey: predicted = true

orange: shifted, unsafe side - the order survives

TALK · 2 MIN How would you decide, in your own words, what a fair test of this screen is? Then, in the drawing above, why does a constant shift keep the ranking - and why was finding the 185 ms a cheat?

CONCLUSION

seen contingency	accurate: 0.69 ms mean error
unseen one	right order, but each CCT 57 ms too long: cases look safer
185 ms	catches every insecure case, but was chosen knowing the answers
in service	a contingency not seen in training is simulated

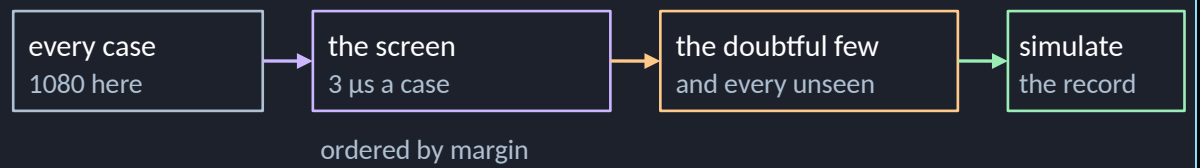
The network does not carry over to a grid it never saw: it can order the cases, not certify them.

Where the Screen Fits

Screening in service — where the trained screen would live and what stays a simulation. At three microseconds a case the network orders the whole queue in milliseconds; the simulation, the tool of record, is spent on the doubtful and the unseen. Ex_12.2 times both.

THE DIVISION OF LABOUR

the screen	every case, every few minutes, locally
the simulation	the doubtful cases, and every unseen topology
the record	always the simulation's, never the screen's
retraining	when the grid itself changes



TIME FOR ONE CASE

method	time
simulation, 2 ms step (reference)	1.7 ms
simulation, 5 ms step	0.6 ms
graph network	3 µs
all 1080 cases, simulated together	1.7 s

On six buses the simulation is already fast. On a real grid of thousands of buses one case takes seconds, and thousands of cases must be checked every few minutes: the network's microseconds decide which cases are simulated first.

Other Ways to Learn the Screen

Alternative formulations — the screening problem can be given to deep learning in more than one shape, and this block has met each of them: a regression of the clearing time, a classification of secure against insecure, and a network that learns the swing itself.

THREE SHAPES, AND WHAT EACH ONE PAYS

the formulation	it returns	it needs	its cost
regress the clearing time - Ex_12.2	a margin, so a ranking	simulated labels	a simulation per training case
classify secure or insecure - L11.2's loss	a verdict at a threshold	the cost of each error	the margin is lost
learn the swing $\delta(t)$ - L12.1's way	the trajectory itself	the physics in the loss	one network per case; hardest
dense network on a feature list	a time, one topology	a fixed grid	cannot take an outage

WHY EX_12.2 CHOSE THE FIRST

Screening needs the margin, and every contingency changes the topology, so the exercise regresses the clearing time with a graph network. One toolbox, three shapes.

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
the dense baseline	what structure wins against, measured	UDL ch. 13	the claim the deck does not make
exciters and governors	control acting within the first swing	ANDES docs	beyond the classical labels
ANDES, a tool of record	the labels checked against a full simulator	ANDES docs	Ex_12.2 section 5
frequency and voltage stability	the two questions not asked here	ENTSO-E	low-inertia and weak grids
one-shot against rollout	the verdict at once, or the states marched	Liu ch. 5	longer horizons

WHY THEY ARE LEFT OUT

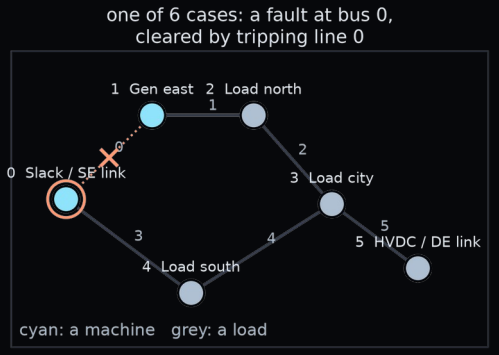
Ex_12.2 screens one grid's 1080 cases against its own Runge–Kutta reference, so none of them is needed to do it. The dense baseline stays an open measurement.

EXERCISE

Exercise 12.2

ONE NOTEBOOK, SECTION BY SECTION

- **1 – 3** — the contingencies, the swing, one case 10 ms either side
- **4** — the reference on all 1080 cases; the step study
- **5** — the worst case in ANDES, classical and detailed
- **6** — the graph layer and the loss; 144 points train, 36 test
- **7** — line 0 held out entirely; the threshold raised until safe



THE QUESTION EACH PART ANSWERS

1 – 4	what is insecure, at what step?
5	does a full simulator agree?
6	can a screen match the labels?
7	what happens off the map?

REPORT QUESTION, TO CONCLUDE

On the unseen outage the screen was 57 ms wrong, on the unsafe side, and still ranked the cases. Write the operating rule you would give a control room for this screen: when is it trusted, and when is it ignored?

SUMMARY

Key Takeaways

TRANSIENT STABILITY	
what it is	do the machines stay in sync?
function	a clearing time per case
train, tune	simulated labels, bisected
vs a verdict	a margin: CCT − 140 ms
used for	the queue of N−1 cases

GRAPH PREDICTOR	
what it is	message passing on the outage graph
function	the clearing time, in 3 µs
train, tune	on simulations; 6721 weights
vs unseen samples	topologies are the fair test
used for	ordering the work, never the record

Next — L13: the mini projects, where one of these problems becomes yours.

Questions

1 – 2 · LEARNING THE CLEARING TIME

1 **Why is a network trained to predict each case's critical clearing time rather than stable or not?**

Every operating point times every outage is a case, and each label costs about ten simulations. A trained network answers in microseconds. The clearing time minus the 140 ms of the protection is a margin, so the cases can be ranked, nearest to the limit first. *Go deeper: the labels and their cost (Ex_12.2 sections 3 – 4); Kundur ch. 13.*

2 **Where is the physics in the graph network, if not in its loss?**

The loss is only the squared error to the simulated clearing time. The physics is in the labels, from the swing equation, and in the graph the network is given. It learns how the clearing time varies on the grids it has seen; it does not solve the swing equation. *Go deeper: the training (Ex_12.2 section 5); UDL ch. 13.*

3 – 4 · THE GRAPH AND THE UNSEEN LINE

3 **How does a graph network take a line outage as input?**

A tripped line is an edge removed from the graph, not a value in a list. Each layer combines every bus's numbers with its neighbours' along the lines that remain, with the same weights everywhere, so any graph can be given. A dense network takes a fixed list. *Go deeper: the graph layer (Ex_12.2 section 5); UDL ch. 13.*

4 **Why is an unseen topology the fair test of the graph network, and what did it show?**

Outages seen in training test interpolation: there it was within 0.6 ms. With line 0 out never shown, it was 57 ms off on the unsafe side and missed all eleven insecure cases. The ranking held, but the threshold that rescued it was found with the truth known. *Go deeper: the unseen line (Ex_12.2 section 6); what that means in a control room.*

NEXT

L13 — The Mini Projects

One month, one problem of your own, chosen from the Part 2 sets: thermal, flow, battery, solid oxide, the car, or the grid - and the Thor runs the screen.

BEFORE THE NEXT SESSION

read	UDL ch. 13
run	Ex_12.2, sections 1 to 6
bring	your operating rule for the screen
choose	the mini project you will take

Part 2 is finished: heat, flow, electrochemistry, a car, and a grid - every one a physics, a classical answer, and a network earning a stated place beside it.